Question 1

a)

- The lower limit of the gate delay angle at the rectifier is dictated by the development of adequate forward voltage to ensure reliable turn of the thyristors
- The lower limit of extinction advance angle at the inverter is dictated by the risk of commutation failure
- The upper limit of gate delay and extinction advance angle for both rectifier and inverter are dictated by their reactive power consumption which goes up with increase of gate delay and extinction advance angle

[1 mark each – 3marks]

b)

- Current control at rectifier (inverter) requires less (high) reactive power at high loading.
- DC fault currents can be automatically limited with current control at the rectifier.
- Voltage control at inverter provides better voltage regulation for loads when in case of HVDC links connecting a generation rich area to a load centre.

[1 mark each – 3 marks]

c)

i) <u>DC line/cable</u>

- Direct voltage reduces, direct current increases at the rectifier end and decreases at the inverter end.
- Rectifier control increases α up to the region of inversion (e.g. $\alpha = 140°$)
- Inverter continues to control γ.

[1 mark each – 3 marks]

ii) <u>Inverter end AC system</u>

- With constant extinction advance angle, the inverter end direct voltage would reduce and the direct current would increase
- Rectifier controls the direct current by increasing the gate delay angle α
- Inverter continues to operate at a fixed γ.

[1 mark each - 3 marks]

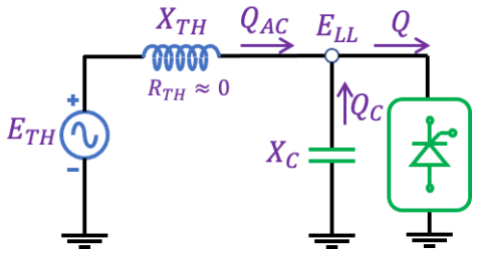
iii) <u>Rectifier end AC system</u>

- Rectifier end direct voltage and hence the direct current would reduce
- Rectifier controls the current by reducing the gate delay angle α
- Once the gate delay angle is stuck at its minimum limit, the inverter takes over current control
- Inverter takes over current control and controls the current by reducing its direct voltage by increasing the extinction advance angle γ

Model Answers

[1 mark each – 4 marks]

d) The AC system represented by its Thevenin equivalent connected to the inverter as shown in the figure.



$$E_{TH} - E_{LL} \approx Q_{AC}X_{TH} = (Q - Q_C)X_{TH} = \frac{Q - Q_C}{\text{SCL}}$$

- If the AC system voltage (E_{TH}) reduces, the converter terminal voltage (E_{LL}) also reduces resulting in a decrease in the reactive power (Q_C) produced by the local capacitor and filter banks as square of the terminal voltage.

- Considering operation at rated power, the deficit in reactive power ($Q - Q_C$) is transmitted from elsewhere in the AC system which incurs voltage drop over the predominantly reactive transmission lines resulting in further reduction of terminal voltage (E_{LL}) potentially triggering a cascaded problem.

- The extent of voltage drop and hence, potential cascading of the above problem is determined by the Thevenin equivalent impedance (X_{TH}).

- If X_{TH} is high (low) which represents a weak (strong) AC system, the LCC-HVDC would therefore be vulnerable to this cascaded problem.

- A weak AC system resulting in larger voltage reduction at the inverter terminal makes it vulnerable to commutation failure.

[1 mark each – 5 marks]

e) Line-to-line voltage

$$e_{ba} = e_b - e_a = \sqrt{3}E_m \cos(\omega t - 90°)$$
$$e_{ab} = e_a - e_b = \sqrt{3}E_m \cos(\omega t + 90°)$$

[1 mark]

$$V_{d0} = \frac{1}{\pi/3}\int_{\pi/3}^{2\pi/3} e_{ba}\,d\theta$$

(Limits should be $-\frac{2\pi}{3}$ to $-\frac{\pi}{3}$ if e_{ab} is used)

Model Answers

[1 mark]

$$= \frac{1}{\pi/3} \int_{\pi/3}^{2\pi/3} \sqrt{3} E_m \cos(\omega t - 90°)\, d\theta$$

$$= \frac{3\sqrt{3}}{\pi} E_m \sin(\omega t - 90°)\Big|_{\pi/3}^{2\pi/3} = \frac{3\sqrt{3}}{\pi} E_m = \frac{3\sqrt{6}}{\pi} E_{LN} = \frac{3\sqrt{2}}{\pi} E_{LL}$$

[2 marks – 4 marks]

Model Answers

Question 2

a) Base case

i)

$B = 2, P_{d-rated} = \frac{2000}{2} = 1000$ MW, $V_{d-rated}$=500 kV, $E_{ac} = 400$ kV, $T = 0.5$

$I_d = \frac{P_{d-rated}}{V_{d-rated}} = 2$ kA

[1 mark]

At inverter end

$V_{doi} = \frac{3\sqrt{2}}{\pi} E_{ac} BT = 540.19$ kV

[1 mark]

$V_{di} = V_{doi} \cos \gamma_{min} - BR_c I_d = 496.59$ kV

[1 mark]

At rectifier end

$V_{dr} = V_{di} + R_L I_d = 501.59$ kV

$V_{dor} = \frac{3\sqrt{2}}{\pi} E_{ac} BT = 540.19$ kV

[1 mark]

$\alpha = \cos^{-1}\left[\frac{V_{dr} + BR_c I_d}{V_{dor}}\right] = 15.08°$

[1 mark – 5 marks]

ii)

Power factor angle at rectifier

$\phi_r = \cos^{-1}\left(\frac{V_{dr}}{V_{dor}}\right) = 0.38$ rad $= 21.79°$

$Q_r = 2 \times V_{dr} I_d \tan \phi_r = 802.16$ MVAr

[1 mark each – 2 marks]

Power factor angle at inverter

$\phi_i = \cos^{-1}\left(\frac{V_{di}}{V_{doi}}\right) = 0.405$ rad $= 23.18°$

$Q_i = 2 \times V_{di} I_d \tan \phi_i = 850.48$ MVAr

Model Answers

[1 mark each – 2 marks]

iii)

$$V_{dr} = \frac{V_{dor}}{2}[\cos\alpha + \cos(\alpha + \mu)]$$

$$\mu_r = \cos^{-1}\left(\frac{2V_{dr}}{V_{dor}} - \cos\alpha\right) - \alpha = 11.86°$$

[1 mark each – 2 marks]

b) Rectifier end voltage reduced to 92%

i)

At rectifier end

$$E'_{acr} = 0.92E_{acr} = 368 \text{ kV}$$

$$V'_{dor} = \frac{3\sqrt{2}}{\pi}E'_{acr}BT = 496.97 \text{ kV}$$

[1 mark]

$$\cos\alpha = \frac{V_{dr} + BR_cI_d}{V'_{dor}} = 1.05 > 1$$

Hence, the rectifier cannot control current any further and the inverter takes over current control leaving the rectifier at α_{min}

[1 mark]

$$I'_d = I_d\left(1 - \frac{I_m}{100}\right) = 1.7 \text{ kA}$$

$$V'_{dr} = V'_{dor}\cos\alpha_{min} - B\,R_cI'_d = 478.76 \text{ kV}$$

[1 mark]

$$V'_{di} = V'_{dr} - R_LI'_d = 474.51 \text{ kV}$$

$$\gamma = \cos^{-1}\left[\frac{V'_{di} + BR_cI'_d}{V_{doi}}\right] = 24.51°$$

[1 mark – 4 marks]

ii)

Power factor angle at rectifier

Model Answers

$$\phi_r' = \cos^{-1}\left(\frac{V_{dr}'}{V_{dor}'}\right) = 0.27 \text{ rad} = 15.56°$$

$$Q_r' = 2 \times V_{dr}' I_d' \tan\phi_r' = 453.22 \text{ MVAr}$$

[1.5 mark]

Power factor angle at inverter

$$\phi_i' = \cos^{-1}\left(\frac{V_{di}'}{V_{doi}}\right) = 0.5 \text{ rad} = 28.55°$$

$$Q_i' = 2 \times V_{di}' I_d' \tan\phi_i' = 877.71 \text{ MVAr}$$

[1.5 mark – 3 marks]

iii)

$$V_{di}' = \frac{V_{doi}}{2}[\cos\gamma + \cos(\gamma + \mu)]$$

$$\mu_i = \cos^{-1}\left(\frac{2V_{di}'}{V_{doi}} - \cos\gamma\right) - \gamma = 8.02°$$

[1 mark each - 2 marks]

c) Inverter end voltage reduced to 94%

$$E_{aci}' = 0.94 E_{aci} = 376 \text{ kV}$$

$$V_{doi}' = \frac{3\sqrt{2}}{\pi} E_{aci}' BT = 507.78 \text{ kV}$$

[1 mark]

$$V_{di}' = V_{doi}' \cos\gamma_{min} - BR_c I_d = 475.59 \text{ kV}$$

$$V_{dr}' = V_{di}' + R_L I_d = 480.59 \text{ kV}$$

[1 mark]

$$\alpha' = \cos^{-1}\left[\frac{V_{dr}' + BR_c I_d}{V_{dor}}\right] = 22.08°$$

[1 mark]

Power factor angle at rectifier

Model Answers

$$\phi_r' = \cos^{-1}\left(\frac{V_{dr}'}{V_{dor}}\right) = 0.47 \text{ rad} = 27.17°$$

$$Q_r' = 2 \times V_{dr}' I_d \tan \phi_r' = 986.61 \text{ MVAr}$$

[1 mark]

Power factor angle at inverter

$$\phi_i' = \cos^{-1}\left(\frac{V_{di}'}{V_{doi}'}\right) = 0.36 \text{ rad} = 20.51°$$

$$Q_i' = 2 \times V_{di}' I_d \tan \phi_i' = 711.64 \text{ MVAr}$$

[1 mark – 5 marks]

Model Answers

Question 3

a)
i)
Zero power implies that there's no current going to the AC grid, this means that the voltage across the inductor is zero, therefore the voltage the converter must apply is V_o=230V in phase with V_{grid}.
The maximum rms AC voltage the DC link can apply is $V_{DC}/\sqrt{2}$. Therefore the minimum DC voltage required is V_{DCmin}=325 V.

[1.5 marks each part – 3 marks total]

Taking the voltage of the grid as the reference of angles, the power delivered to the grid can be calculated as: $\boldsymbol{P_{grid} = V_{grid} \cdot \Re e\{\bar{I}_o\}}$ and $\boldsymbol{Q_{grid} = -V_{grid} \cdot \Im m\{\bar{I}_o\}}$. Therefore, in this case $\boldsymbol{\bar{I}_o = I_o = \frac{P_{rated}}{V_{grid}} = 6.5\ A}$.

The reactive power consumed by the filter can be calculated as: $\boldsymbol{Q_{filter} = X(I_o)^2} = 134$ VAr

The losses due to the resistance of the circuit are: $\boldsymbol{P_{loss} = R(I_o)^2}$ =19 W

[1 mark each part – 3 marks total]

This question is about the frequency response of the DC link and how the 100 Hz current coming from the bridge affects the voltage. The 100 Hz current will flow through the capacitor of the DC link. Some students will work with the frequency response at 100 Hz starting from the transfer function that is used to design the voltage controller which relates voltage and current across the capacitor.

Alternatively, they may remember that this is the definition of the impedance of the capacitor at 100 Hz. Which is $X_C = -\frac{1}{2\pi 100\ C}$.

[2 marks for formulating the link between voltage and current at 100 Hz either way]

The impedance relates voltage and current: $\bar{V}_C = -jX_C\bar{I}_C$. From the specification of the problem we know the voltage (4V) and the current (3.75A) and we can solve for the impedance: $X_c \leq \frac{V_C}{I_C}$. This equation can be used with the RMS value or with the peak-to-peak value as long as there's consistency between voltages and currents. Therefore we get X_C= 1.07 Ω which gives C= 1.5mF.

[2 marks for solving for the minimum capacitance]

ii)

This question can be answered by looking at the frequency response of the closed-loop system at the frequency of 50 Hz. The frequency response can be obtained as:

$\boldsymbol{\frac{I_o}{I_o^*}(\omega) = \frac{1}{1+j\omega\tau}}$ which at 50 Hz becomes: $\boldsymbol{\frac{I_o}{I_o^*}(2\pi 50) = \frac{1}{1+j\tau 2\pi 50}}$ for a 1 A reference, the output current has an amplitude of $\boldsymbol{\left\|\frac{1}{1+j\tau 2\pi 50}\right\| = \frac{1}{1+(\tau 2\pi 50)^2}}$ =0.53 A. The angle delay is -43.3º.
[3 marks]

iii)
The equation that relates the voltage of the capacitor to its current is $\frac{d}{dt}V_{DC} = \frac{1}{C}I_{DCc} = \frac{1}{C}(I_{DCi} - I_{DCo})$.

Model Answers

Once transformed by Laplace this gives: $V_{DC}(s) = \frac{1}{Cs} I_{DCi}(s) - \frac{1}{Cs} I_{DCo}(s)$

[1 marks]

b)

i)

We are told that the power delivered to the AC grid is 50 MW at unity power factor. We can calculate the required AC current as $I_{AC}=P/V_{AC}$=1.32 kA. Next we can calculate the voltage the stacks must apply in order to feed this current. This can be solved using the equivalent transformed circuits for the 2-arm MMC, ie solving for the differential and summative variables or directly working with upper arm and lower arm variables. We know that this corresponds to a stable operating point with no drift of the energy of the stacks. We also know that the voltages of the DC link are balanced. This implies that the converter shouldn't use AC current internally to ship power between the upper stack and the lower stack, which implies that the AC current going through each of the stacks is exactly one half of the AC grid current. Ie we know that $I_{uAC}=I_{AC}/2$ and $I_{lAC}=-I_{AC}/2$. Therefore, we can write the following two equations:

$$-V_{AC} - \bar{V}_u^{AC} = (R_L + jX_L)I_{AC} + (R_a + jX_a)I_{AC}/2$$
$$-V_{AC} + \bar{V}_l^{AC} = (R_L + jX_L)I_{AC} + (R_a + jX_a)I_{AC}/2$$

Solving for the AC component of the output voltage of the upper and lower stacks we get: V_{uAC}=38.3 kV with angle -172º and V_{lAC}=38.3 kV with angle 8.3º.

Knowing the AC voltage applied by the stacks and their AC current we can find the active power they exchange and find the power that needs to be drawn by the stacks using DC current.

$$P_{u,l}^{AC} = \Re e\{\bar{V}_{u,l} \cdot \bar{I}_{u,l}^*\}$$

Note that the sign is set according to the direction of the current and voltage in the diagram. Ie, a positive value would mean that the stacks absorb power. This gives: $P_{uAC}=P_{lAC}$=-25.2 MW

The power exchanged using DC would be the same as the power exchanged with AC but with the opposite sign. We don't know the DC voltage or the DC current, this leads to a quadratic equation. However, if we assume the losses of the resistance to be very small (which we can confirm later) we can calculate the DC current assuming that the voltage drop due to the resistance of the circuit is zero:

$I_{uDC}=2\,P_{uDC}/V_{DC}$ and $I_{lDC}=2\,P_{lDC}/V_{DC}$. This gives $I_{uDC}=I_{lDC}$=420 A. This assumes that the voltage applied by the stacks is $V_{uDC}=V_{DC}/2$ and $V_{lDC}=V_{DC}/2$, which is 60 kV. We can confirm if the assumption is reasonable by calculating the DC voltage drop due to the resistance. This gives: $V_{RDC}=2\,R_a\,I_{DC}$=235 V, which is 0.2% of V_{DC}.

[2 mark for calculating IAC, 2 marks for finding the AC components of the voltage and the currents of the stacks, 2 marks for calculating the DC components]

ii)

This question requires the students to remember the definition of the summative and differential variables:

$I_{sum} \triangleq \frac{1}{2} I_u + \frac{1}{2} I_l$ and $I_{AC} = I_u - I_a$.

This gives:

$\bar{I}_{sum}^{AC}$ =7.5 A

Model Answers

I_{sum}^{DC} =25 A

$\bar{I}_{AC}^{AC}$ =35 A

$\boldsymbol{I_{AC}^{DC}}$ =-20 A

[2 marks]

iii)

The summative current is a current that flows between the HVDC link and the converter legs without flowing through the AC grid. The DC component more or less interacts identically with the upper stack and the lower stack and it is used to ship power between the HVDC link and the legs to compensate for the power delivered to the AC grid. The AC component interacts between the upper stack and the lower stack with opposite signs; therefore it is used to ship power from the upper stack to the lower stack.

The discussion is about the single-phase 2-arm MMC, therefore no discussion of the implications in the case of three-phase are expected.

[2 marks]

iv)

This question ask us to derive the equations of the circuit. This can be done by analyzing the circuit diagram. We can put I_{AC} as a function of I_u and I_l directly:

$$-V_{AC}(t) - V_u(t) + \frac{V_{DC}(t)}{2} = R_a I_u(t) + L_a \frac{d}{dt} I_u(t) + R_L\big(I_u(t) - I_l(t)\big) + L_l \frac{d}{dt}\big(I_u(t) - I_l(t)\big)$$

and

$$-V_{AC}(t) + V_l(t) - \frac{V_{DC}(t)}{2} = -R_a I_l(t) - L_a \frac{d}{dt} I_l(t) + R_L\big(I_u(t) - I_l(t)\big) + L_l \frac{d}{dt}\big(I_u(t) - I_l(t)\big)$$

[1 mark]

From observing the equations, it is clear that there's coupling between upper voltages and lower currents.

[1 mark]

v)

The current of the two poles of the HVDC link except for the difference caused by the current flowing through the middle point, which flows back to the circuit through the AC grid. Under normal circumstances, the only current flowing through the AC grid is AC current, which doesn't interact with the voltage of the HVDC poles (which is purely DC). Consequently, the DC current flowing the two poles is identical. If the voltages of the two poles are identical, the power the two poles exchange is identical. If voltages were different, the power exchange of the two poles would be different, but this isn't what the question asks (the circuit shows that the voltage of the two poles is identical). The other way to make the power exchange asymmetrical would be to make the current of the two poles be different.

Model Answers

[1 mark will be awarded if a good explanation of how the current of the two DC poles can be different is provided]

This would require DC current to flow through the middle point and back to the circuit through the AC grid. Typically this isn't acceptable because the DC current flowing through the transformer would cause saturation of the transformer (not represented in the circuit) and all circuit elements on the AC side of the converter.

[1 mark for arguing why it isn't a bad idea to make DC current flow through the AC side of the converter]

vi)

Half-bridge submodules can only output voltages between 0 and their capacitor voltage. In constrast, full-bridge submodules can generate voltages between positive and negative sign capacitor voltage. In the case of half-bridges this constraints the rated AC voltage; it has to be low enough to avoid crossing zero when subtracted from the DC voltage the stacks must apply. This also causes half-bridge submodules to lose control over the current if the DC voltage collapses; which requires a circuit breaker to interrupt the fault current.

[1 mark for discussing the usefulness of being able to apply negative voltage, half mark subtracted if they note that the voltage can be negative but they don't explain that this enables interrupting DC fault current]

Full-bridge submodules can apply negative voltages, which means they can retain control of the current even of the DC voltage is zero. The caveat is that full bridge submodules have twice as many power modules in the current path, which makes their conduction losses (the main losses) be about twice the losses they would have if they were half-bridge submodules. It is possible to combine full bridge and half bridge submodules in a converter to retain the ability to stop fault current while keeping the conduction losses small.

[1 mark for explaining that conduction losses can double because two power modules are in the current path, half mark awarded if they miss the important point about the conduction losses but note that reliability might be a concern (more devices, more chances of problems even if the comparison isn't obvious)]

Model Answers